Discussion Exercise 2

University of California, Riverside
June 29, 2023

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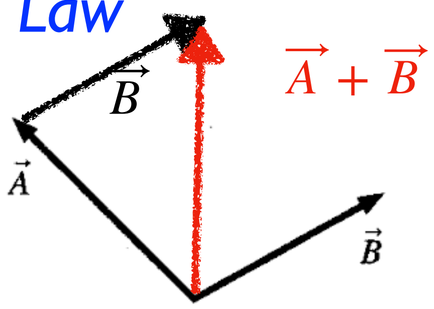
1. Addition and Subtraction of Vectors

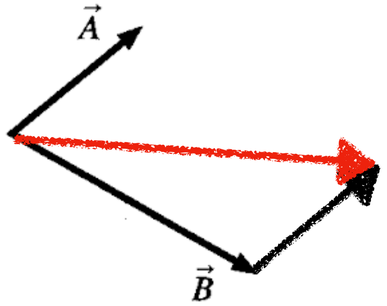
Exercises 1–3: Draw and label the vector sum $\vec{A} + \vec{B}$.

Triangle Law

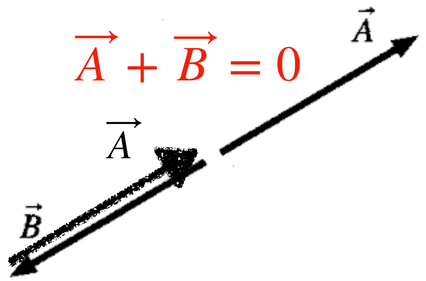
1.



2.

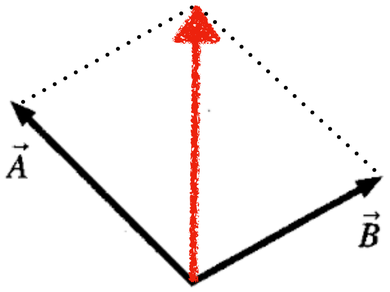


3.



Parallelogram Law

1.



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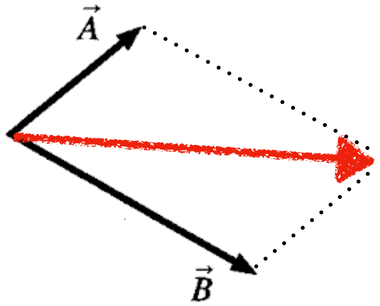
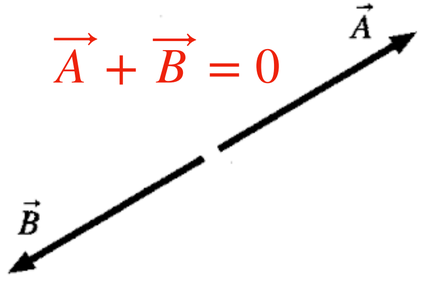


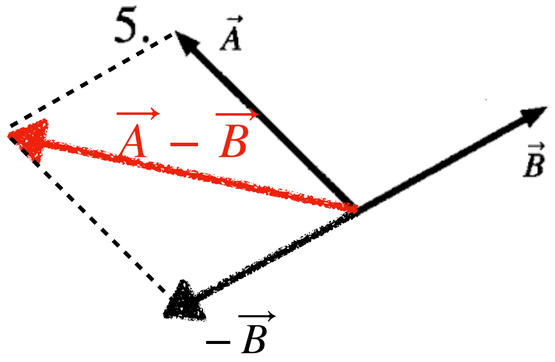
3.



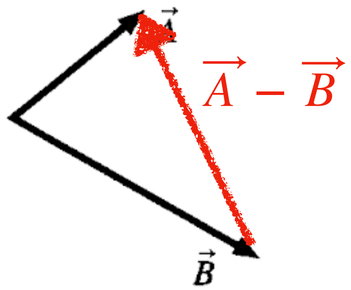
1. Addition and Subtraction of Vectors

Exercises 5–7: Draw and label the vector difference $\vec{A} - \vec{B}$.

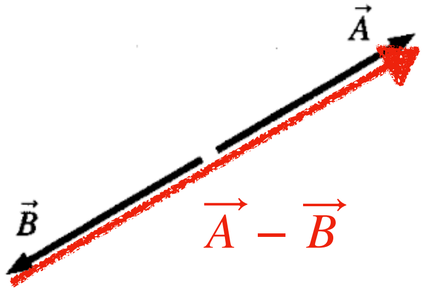
5.



6.



7.

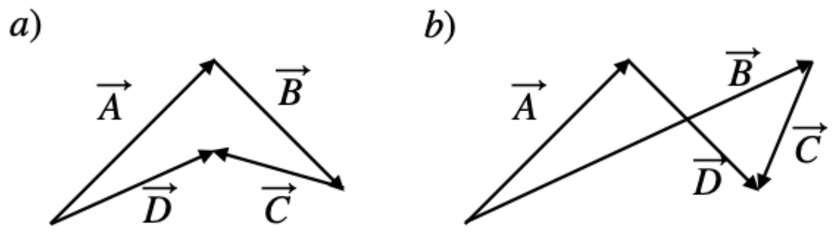


Note: *One can firstly determine '-B', which represents a vector with the same magnitude as B but points in the opposite direction. Then we can apply the rules of addition of vectors.*

Or one can direction draw a vector from the head of B to the head of A. (Why?)

1. Addition and Subtraction of Vectors

2. Write a vector equation for **A** for each arrangement of vectors.



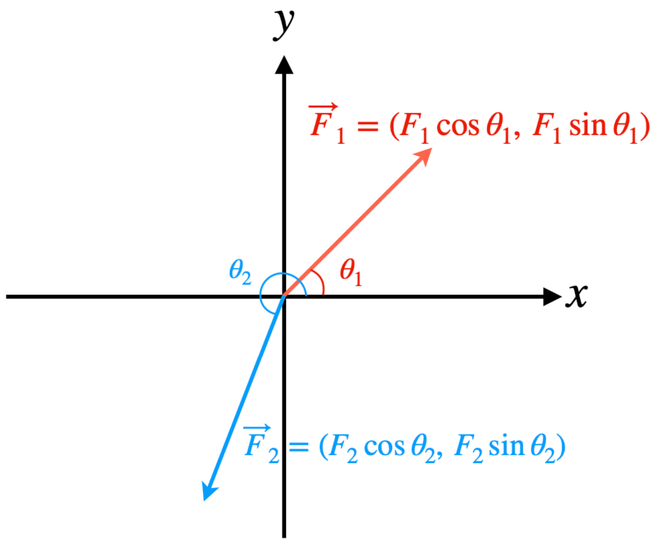
Answer:

a) It is helpful to choose a positive direction when vectors form a closed loop. For example, we can choose the clockwise direction to be positive. If a vector goes clockwise along this loop, then assign a positive sign to it, otherwise assign a negative sign. $A + B + C - D = 0 \Rightarrow \mathbf{A} = -\mathbf{B} - \mathbf{C} + \mathbf{D}$

b) $\mathbf{A} + \mathbf{D} - \mathbf{C} - \mathbf{B} = 0 \Rightarrow \mathbf{A} = \mathbf{B} + \mathbf{C} - \mathbf{D}$

2. Vector Components

[General Method for $\vec{T} = \vec{F}_1 + \vec{F}_2$]



$$T_x = F_1 \cos\theta_1 + F_2 \cos\theta_2 \quad (1)$$

$$T_y = F_1 \sin\theta_1 + F_2 \sin\theta_2 \quad (2)$$

$$T = \sqrt{T_x^2 + T_y^2} \quad (3)$$

$$\theta_{\vec{T}} = \arctan(Ty/Tx) \quad (4)$$

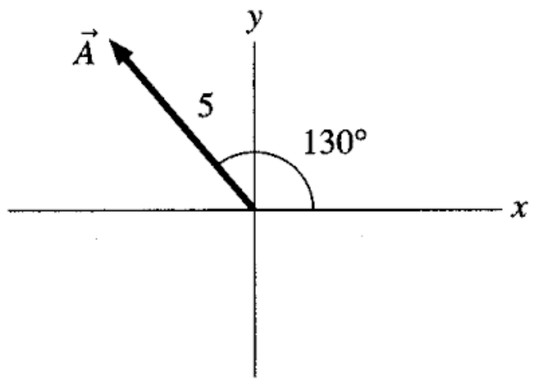
Caveats:

1. Be careful to Eq(4)! $y = \tan(\theta)$ is a periodic function with period π. For a given y, there are two solutions of θ within $[0, 2\pi]$. If your calculator gives a result $\theta \mathrm{mod} 2\pi$, then $(\theta + \pi)\mathrm{mod} 2\pi$ is also a solution, so you need to determine which one is the physical solution.
2. Pay attention to the unit of angle in your calculator! Radian(rad) or degree(°)?

2. Vector Components (Vectors→Components)

Exercises 15–17: Determine the numerical values of the x- and y-components of each vector.

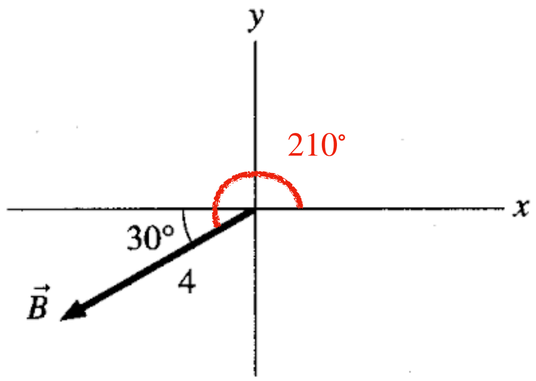
15.



$A_x =$ 5 cos(130°) = – 3.21

$A_y =$ 5 sin(130°) = 3.83

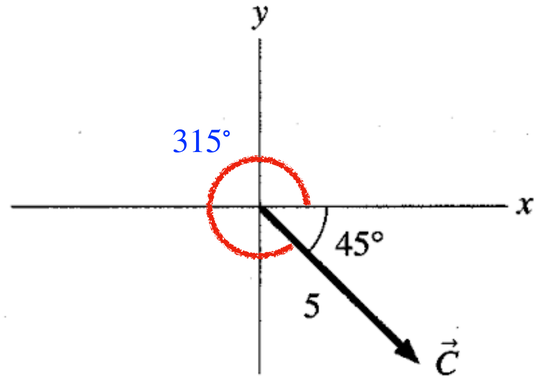
16.



$B_x =$ 4 cos(210°) = – 3.46

$B_y =$ 4 sin(210°) = – 2

17.



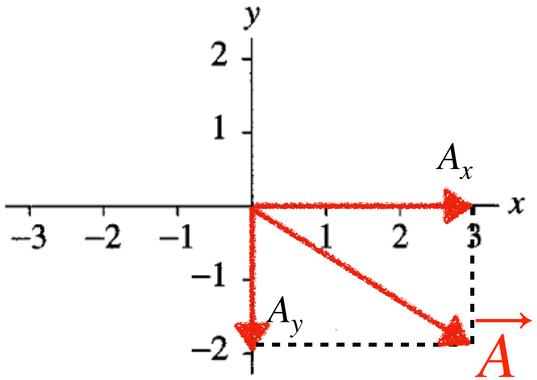
$C_x =$ 5 cos(315°) = 3.54

$C_y =$ 5 sin(315°) = – 3.54

2. Vector Components (Components→Vectors)

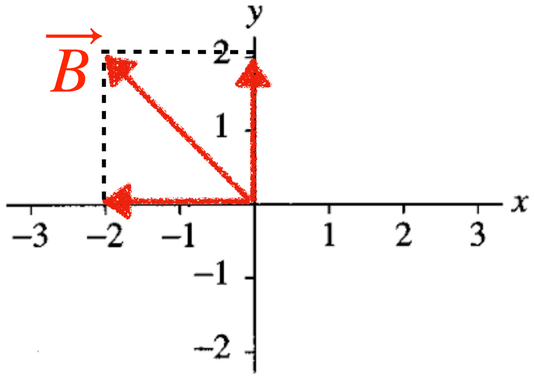
Exercises 18–20: Draw and label the vector with these components. Then determine the magnitude of the vector.

18. $A_x = 3, A_y = -2$



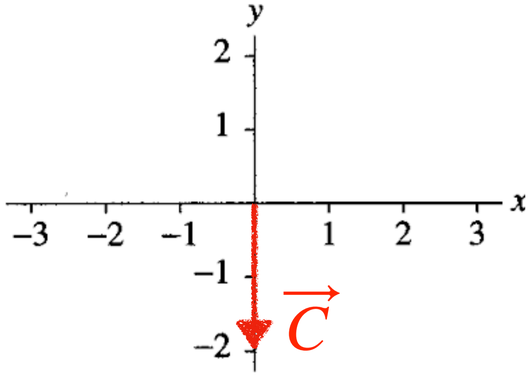
$A =$ $\sqrt{3^2 + (-2)^2} = \sqrt{13}$

19. $B_x = -2, B_y = 2$



$B =$ $2\sqrt{2}$

20. $C_x = 0, C_y = -2$



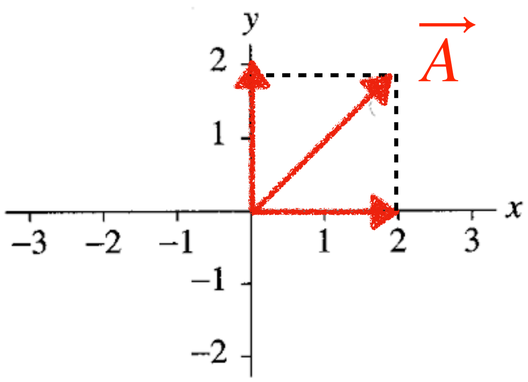
$C =$ 2

2. Vector Components

Exercises 32–34: For each vector:

- Draw the vector on the axes provided.
- Draw and label an angle θ to describe the direction of the vector.
- Find the magnitude and the angle of the vector.

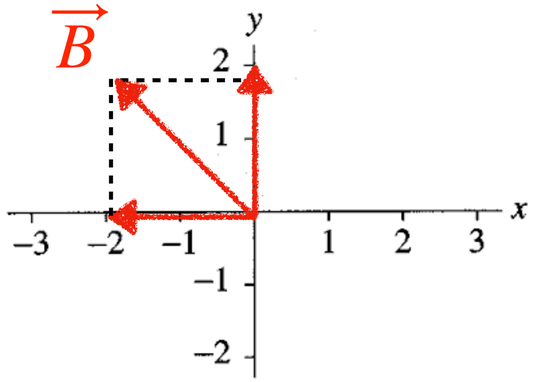
32. $\vec{A} = 2\hat{\imath} + 2\hat{\jmath}$



$A =$ $2\sqrt{2}$

$\theta =$ $45°$

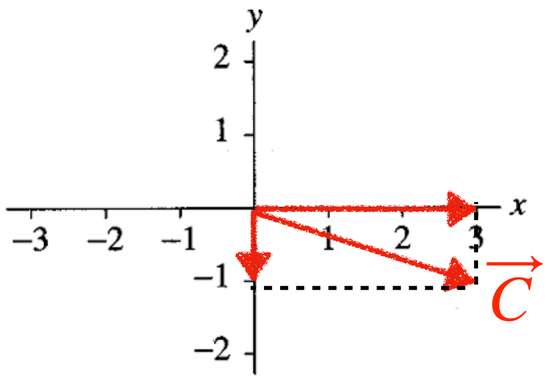
33. $\vec{B} = -2\hat{\imath} + 2\hat{\jmath}$



$B =$ $2\sqrt{2}$

$\theta =$ $135°$

34. $\vec{C} = 3\hat{\imath} - \hat{\jmath}$

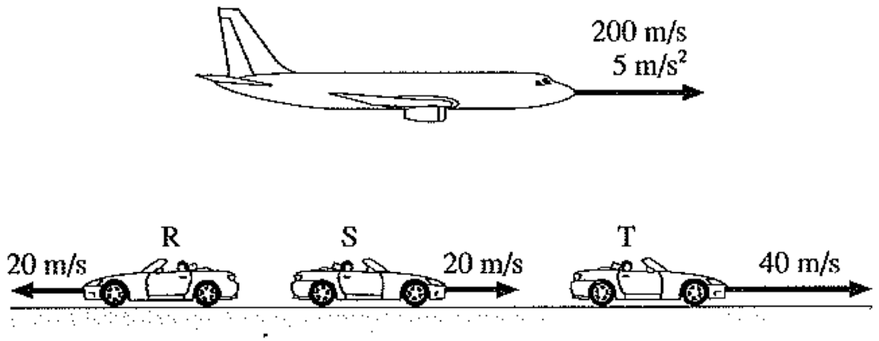


$C =$ $\sqrt{10}$

$\theta =$ $-18.4°$

3. Relative Motion

Ryan, Samantha and Tomas are driving their convertibles. At the same instant, they each see a jet plane with an instantaneous velocity of $200\ m/s$ *and an acceleration of* $5\ m/s^2$. *Rank, in order, from largest to smallest, the jet's speed,* v_R, v_S *and* v_T *according to Ryan, Samantha, and Tomas. Explain.*



Note: speed is a scalar, while velocity is a vector

$$V_A > V_T > V_R = V_S$$

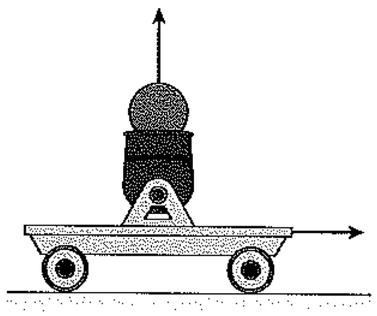
4. Ballistic Cart

A cart rolling at a constant velocity fires a ball straight up. When the ball comes down, will it land in front for the cart, behind the cart, or directly on the cart? Explain.

Land directly on the cart.

The relative horizontal velocity of the ball wrt the car stays zero.

Will your answer change if the cart is accelerating in the forward direction? If so, how?



Land behind the car.

Initially, the relative horizontal velocity of the ball wrt the cart is zero. However, the relative horizontal acceleration of the ball wrt the cart points to the left (since the cart is accelerating in the forward direction), which pulls the ball backward wrt the cart.

5. Circular Motion

$$a = \frac{v^2}{r}$$

An object travels in a circle of radius r at a constant speed v.

Initially, the acceleration is $a = v^2/r$.

1. By what factor does the object's acceleration change if its speed is doubled and the radius is unchanged?

$v' = 2v$, $r' = r$, then $a' = (v')^2/r' = (2v)^2/r = 4v^2/r = 4a$

2. By what factor does the acceleration change if the radius of the circle is doubled and its speed is unchanged?

$v' = v$, $r' = 2r$, then $a' = (v')^2/r' = v^2/(2r) = a/2$

3. By what factor does the acceleration change if the period of motion is doubled without changing the size of the circle?

Recall that $v = \frac{2\pi r}{T}$, then $v' = v/2$, and $a' = \frac{(v/2)^2}{r} = \frac{v^2}{4r} = a/4$